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Chennai, Tamil Nadu

Technical Skills

Programming Languages:

- Python
- R
- SQL
- LaTeX
- Java
- SAS

Data Handling, Analysis, and Visualization:

- pandas
- NumPy
- Matplotlib
- Scikit-learn
- PyTorch
- TensorFlow
- OpenCV
- NLTK
- spaCy
- PowerBI

Teaching & Mentorship

- Teaching Associate
Agentic Systems and Design
IIT Roorkee Jan 2026 - Present
- Teaching Assistant
Minor in AI
IIT Ropar Nov 2024 - Nov 2025
- Teaching Assistant
CMI Aug 2024 - Apr 2025
Assisting Prof. Sourish Das & Prof. Kavita Sutar in teaching courses like, Probability and Statistics with R, Visualisation and Linear Algebra and its Applications.

Invited Talks

- Probabilistic Modeling – Invited Seminar, National Mathematics Day Department of Statistics, Maulana Azad College
Slides Dec 2025
- Conformal Prediction: A Framework for Reliable Uncertainty Quantification – Top 6 Speakers The AI Hour, Vizura AI Labs
Slides Dec 2024
- Data-driven Graph Construction: Unlocking Insights through Clustering and Spectral Partitioning – Invited Speaker Data Science Seminar, CMI
Slides Sep 2024
- AI for Everyone: Bridging the gap between Data and Decision-Making – Invited Speaker Department of Statistics, Maulana Azad College
Slides Aug 2024

Soumyajoy Kundu

DATA SCIENTIST

Profile

Aspiring researcher with a strong foundation in statistics and machine learning, and hands-on experience in Python, R, and SQL. Skilled in analyzing complex data, building models, and deriving actionable insights. Experienced in research-driven projects, with a focus on solving real-world problems. Keen to contribute to cutting-edge AI research and grow through collaboration.

Education

Chennai Mathematical Institute	Master of Science
Data Science Aug 2023 – Jul 2025	CGPA : 9.38 / 10.00
University of Calcutta	Bachelor of Science
Statistics (Honours) Oct 2020 – Jul 2023	CGPA : 9.163 / 10.000

Work Experience

- Datakulture, Sedin Technologies, Chennai Jun 2025 – Present
Junior Data Scientist
 - OmniTale: Developing a multimodal AI pipeline integrating vision, audio, and language models for context-aware reasoning and structured decision-support outputs.
 - Viztran: Developing a vision-language pipeline to interpret dashboard images and map chart elements into structured representations compatible with Power BI, exploring LLM-guided visual parsing using GPT-4o.
 - InvoVision: Developed an AI system for medical invoice understanding using OCR with vision-language models, exploring LLaVA, Qwen-VL, etc., to extract structured fields under layout variability and noisy clinical data.
 - Warehouse Optimization: Formulating a discrete optimization problem for warehouse slotting and inventory placement, focusing on combinatorial optimization, constraint modeling, and cost-efficient allocation under capacity and operational constraints.
- ForeML Lab, Abu Dhabi Sep 2025 – Present
Research Assistant, Sorbonne University and Sorbonne Cluster for AI
 - Developing robust demand forecasting models in diffusion-driven settings by extending Bass-type frameworks with stochastic perturbations and structural uncertainty.
 - Evaluating statistical and ML methods for parameter stability and long-horizon forecasting under the guidance of Dr. Tanujit Chakraborty.
- Theia Technologies, Bangalore Jan 2025 – Apr 2025
Research Intern
 - Designed and deployed a real-time object tracking system using SAM2 and SAMURAI, integrating PTZ camera control on RTSP live streams.
 - Enhanced tracking robustness under rapid object/camera motion via jitter smoothing and system-level performance optimizations.
- Reserve Bank of India, Chennai May 2024 – Jul 2024
Research Associate, Department of Statistics and Information Management
 - Conducted empirical analysis of RBI liquidity management's impact on USD/INR exchange rates using VAR, ARDL, ECM, and GARCH models with curated central bank data.
 - Evaluated model performance, demonstrating stronger reliability of econometric approaches over ML models (RNN, LSTM) for small-sample time series forecasting.

Academic Achievements

- 1. CSIR NET (JRF) Dec 2025
AIR 18 2025
- 2. GATE DS & AI (DA) 2025
AIR 129 2025
- 3. GATE Statistics (ST) 2025
AIR 03 2025
- 4. IIT JAM MS 2024
AIR 11 2024
- 5. ISI M. Stat 2024
AIR 27 2024
- 6. IIT JAM MS 2023
AIR 100 2023
- 7. ISI M. Stat 2023
AIR 34 2023
- 8. CUET PG 2023 (Statistics)
AIR 01 2023
- 9. University Topper in Statistics
University of Calcutta 2023
- 10. Outstanding Achievement Award
Maulana Azad College, IQAC 2023

Key Courses Taken

- 1. Linear Algebra and its Applications
- 2. Mathematical Methods & Analysis
- 3. Data Mining & Machine Learning
- 4. Deep Learning
- 5. Computer Vision
- 6. Natural Language Processing
- 7. Reinforcement Learning
- 8. Probability and Statistics
- 9. Statistical Inference
- 10. Linear Models and Regression
- 11. Stochastic Processes
- 12. Time Series and Survival Analysis

Articles

- 1. Ozempic, Mounjaro, and other GLP-1 Drugs: How Effective Are They for Weight Loss? 
Datakulture, Sedin Technologies
 - Co-authored a data-driven analysis of Semaglutide and Tirzepatide, evaluating weight-loss efficacy and benefit-risk trade-offs across patient cohorts and dosages using clinical and real-world evidence.

Projects

- 1. GeomVLM: A Lightweight Vision–Language Model for Structured Image – Text Alignment 
Sedin Technologies | Self Dec 2025
 - Designed a compact **Vision–Language Model (VLM)** combining a CNN-based image encoder with a Transformer text encoder for multimodal representation learning.
 - Trained using **CLIP**-style contrastive objectives to learn a shared embedding space for accurate image–text alignment on structured datasets.
- 2. Reprojection, Reconstruction, Realization: Multi-View 3D Geometry via Bundle Adjustment 
Chennai Mathematical Institute | Data Science Feb 2025 – Apr 2025
 - Built a complete **Structure-from-Motion (SfM)** pipeline including feature extraction, pose estimation, triangulation, and bundle adjustment.
 - Implemented **Levenberg–Marquardt** optimization to jointly refine camera parameters and 3D structure by minimizing reprojection error.
- 3. Optimizing Acrobot Dynamics with Deep Reinforcement Learning. 
Chennai Mathematical Institute | Data Science Nov 2024 – Dec 2024
 - Modeled the Acrobot control problem as a **Markov Decision Process** and trained a **Deep Q-Network (DQN)** for optimal policy learning.
 - Performed systematic **hyperparameter tuning** (learning rate, epsilon decay, discount factor) to achieve stable training and consistent reward convergence.
- 4. SonicShield: AI-Powered Guardian Against DeepFake Speech 
Chennai Mathematical Institute | Data Science Mar 2025 – May 2025
 - Designed an **audio anomaly detection framework** using spectrogram-based representations and deep learning for detecting synthetic vs. real speech signals.
 - Evaluated **Transformer-based models (DistilBERT)** for sequence classification, analyzing robustness under noisy inputs and class imbalance in real-world audio settings.

Conferences and Workshops

- 1. International Conference on Applied AI and Scientific Machine Learning (*CASML*), Indian Institute of Science, Bangalore Dec 2025
- 2. Winter School on Deep Learning: Reinforcing Science with Foundation Models (*WSDL*), Electronics and Communication Sciences Unit, Indian Statistical Institute, Kolkata Jan 2025 – Mar 2025
- 3. 4th International Conference on Mathematical, Engineering and Management Sciences Sep 2024
 - Organized by *NIT Meghalaya* and *NIT Patna*.
 - Presented one of my works titled, "*An Analysis of Consumer Behavior in Digital Era Understanding the Shift in Lifestyle and E-Commerce Preferences*".
- 4. Conference for Statistics and Programming in Clinical Research Sep 2024
 - Organized by Indian Association of Statistics in Clinical Trials (*IASCT*).
 - Presented a poster for one of my works titled, "*Data-driven Graph Construction: Unlocking Insights through Clustering and Spectral Partitioning*".